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24. Write a Python program to draw line charts of the financial data of Alphabet Inc. between October 3, 2016 to October 7, 2016.

Aim

To draw line charts for the Open, High, Low and Close financial values of Alphabet Inc. using data stored in a CSV file.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

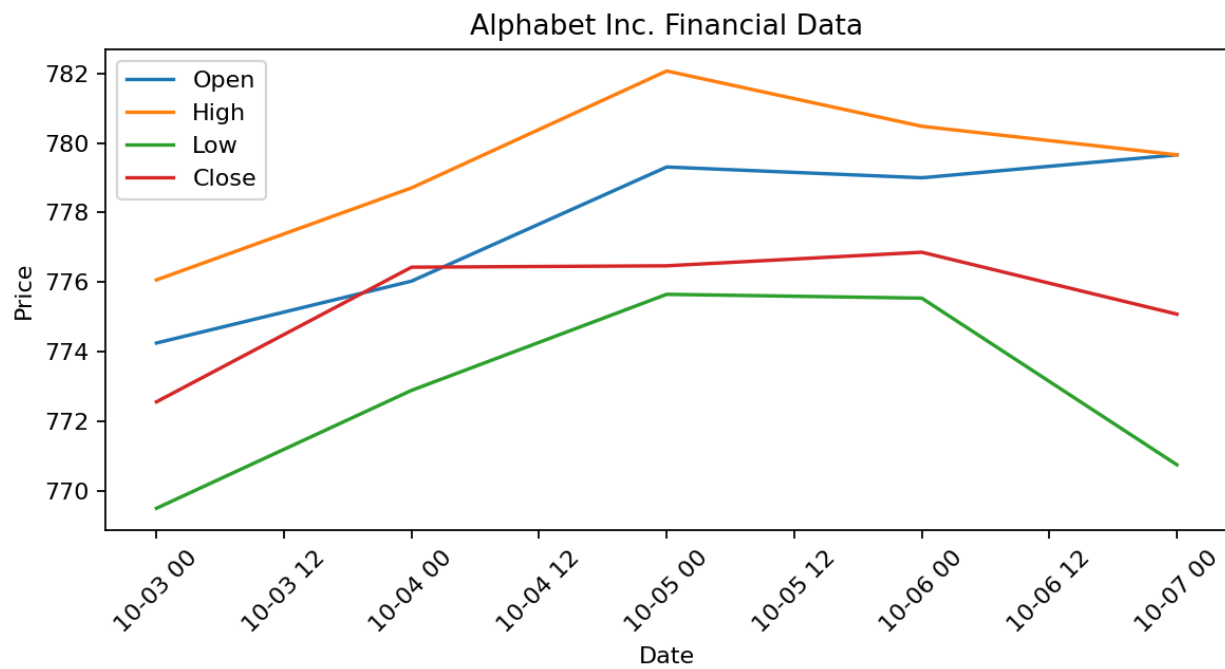
df = pd.read_csv(
    r"C:\Users\Tilak\OneDrive\Documents\fddata.csv",
    encoding="utf-8-sig"
)

df.columns = df.columns.str.strip()
df["Date"] = pd.to_datetime(df["Date"])

plt.plot(df["Date"], df["Open"], label="Open")
plt.plot(df["Date"], df["High"], label="High")
plt.plot(df["Date"], df["Low"], label="Low")
plt.plot(df["Date"], df["Close"], label="Close")

plt.xlabel("Date")
plt.ylabel("Price")
plt.title("Alphabet Inc. Financial Data")
plt.legend()
plt.xticks(rotation=45)
plt.show()
```

Output



25. Write a Python program to plot two or more lines with legends, different widths and colors.

Aim

To plot multiple lines with different colors, line widths and legends using Matplotlib.

Code

```
import matplotlib.pyplot as plt

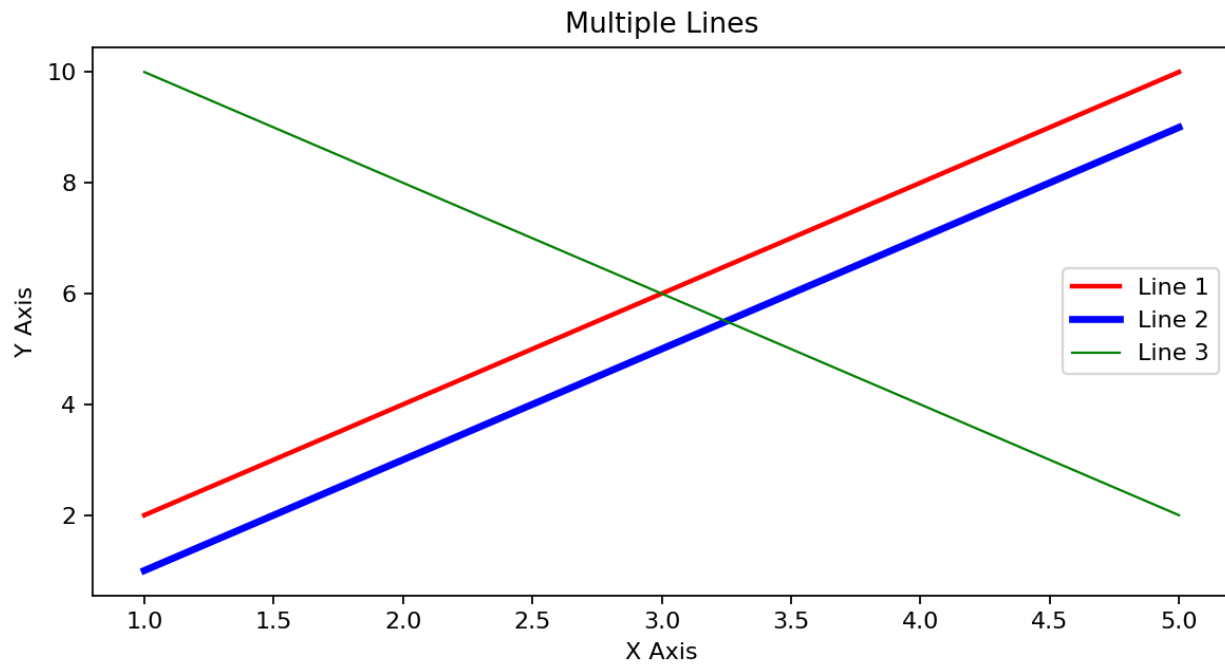
x = [1, 2, 3, 4, 5]

y1 = [2, 4, 6, 8, 10]
y2 = [1, 3, 5, 7, 9]
y3 = [10, 8, 6, 4, 2]

plt.plot(x, y1, color="red", linewidth=2, label="Line 1")
plt.plot(x, y2, color="blue", linewidth=3, label="Line 2")
plt.plot(x, y3, color="green", linewidth=1, label="Line 3")

plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Multiple Lines")
plt.legend()
plt.show()
```

Output



26. Write a Python program to create multiple plots.

Aim

To create and display multiple plots in a single figure using Matplotlib.

Code

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]

y1 = [2, 4, 6, 8, 10]
y2 = [10, 8, 6, 4, 2]

plt.subplot(2, 1, 1)
plt.plot(x, y1)
plt.title("Increasing Line")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")

plt.subplot(2, 1, 2)
plt.plot(x, y2)
plt.title("Decreasing Line")
plt.xlabel("X Axis")
```

```
plt.ylabel("Y Axis")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output

